

feynsage

Feynman Integrals in SageMath, from Graphs to Numbers

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Built on the notes of Prof. B. Ananthanarayan's lecture "Feynman integrals at one loop", NISER, 29 September 2026

github.com/aburousan/feynsage · MIT licence · every feynsage result in this talk was computed by the package

Why another package?

Today a one-loop or multi-loop calculation usually means several programs:

- Package-X, FeynCalc and LiteRed need Mathematica; FIRE uses it for set-up around a C++ reducer
- LoopTools gives numbers, not formulas
- Kira and FIRE reduce, but do not draw or explain

We wanted **one** free tool where every step is exact and can be checked.

FEYNBSAGE

- written natively for SageMath (free, Python)
- exact algebra in $\mathbb{Q}(D, s, t, m, \dots)$ and graph theory come built in
- one loop like Package-X, many loops like LiteRed / Kira
- FORM for the Dirac algebra
- every function explains its output: `explain=True`

What is inside

Step	feynsage	Plays the role of
diagram	<code>graph("A-B, A-B:m", legs, kin), diagram("kite"), pictures</code>	pen and paper
$\mathcal{U}, \mathcal{F}$	spanning trees, 2-forests, Kirchhoff, $\det M$	sir's notebook, FeynCalc
family, sectors	<code>family([...])</code> , zero sectors, symmetries	LiteRed
IBP	Laporta; finite fields + rational reconstruction	FIRE, Kira
one loop	<code>loop(...)</code> $\rightarrow A_0, B_0, C_0, D_0$	Package-X, FeynCalc
values	closed forms of C_0, D_0 ; IR poles	Package-X, LoopTools
Dirac traces	<code>form.dirac_trace(...)</code> , γ_5 included	FORM, FeynCalc

Three lines to start

```
from feynsage import *
loop("l^mu l^nu", ["l", "m"], ["l + p", "m"], kin={"p^2": "s"})      # one loop, exact
ibp_reduce(family(["l1", "l1 + q", "l1 + l2", "l1 + l2 + q", "l2"], kin={"q^2": 1}),
            ["F(2,2,1,2,2)"])      # two loops, IBP
diagram("kite").plot()      # a picture
```

Install: `git clone ... && ./install.sh` (macOS, Linux, WSL; it finds or installs Sage and FORM).

NOTATION, AS IN THE LECTURE NOTE

$D = 4 - 2\varepsilon$, loop momentum ℓ , Feynman parameters x_i , graph polynomials $\mathcal{U}$ and $\mathcal{F} = \mathcal{F}_0 + \mathcal{U} \sum_i x_i m_i^2$. One-loop functions in the Package-X normalisation $\mu^{2\varepsilon} e^{\varepsilon\gamma_E} \int d^D \ell / (i\pi^{D/2})$.

$\mathcal{U}$ and $\mathcal{F}$: the idea of the lecture

Sir's notebook (uf-new-short.nb) completes the square, loop by loop:

$$\sum_i x_i D_i = \sum_{r,s} \ell_r \cdot \ell_s M_{rs} - 2 \sum_r \ell_r \cdot Q_r + J$$

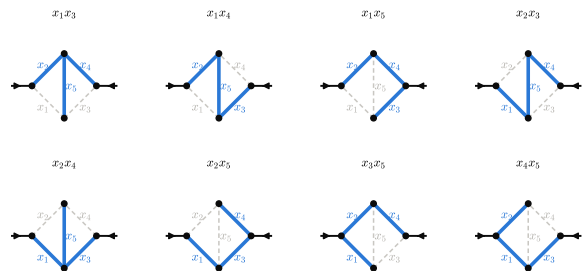
$$\mathcal{U} = \det M, \quad \mathcal{F} = \det M (J - Q^T M^{-1} Q)$$

and gives four examples. feynsage reproduces all four, **two ways**:

diagram	from propagators	from the graph
massless triangle	✓	✓
bubble, two masses	✓	✓
massless box	✓	✓
two-loop sunset	✓	✓

READ OFF THE GRAPH

- $\mathcal{U}$: cut L lines so that a spanning tree is left, multiply their x_i
- $\mathcal{F}_0$: cut $L + 1$ lines so that two trees are left, times the momentum² flowing between them
- $\mathcal{U}$ again: det of the Kirchhoff matrix (matrix-tree theorem)



the 8 spanning trees of the kite, drawn by feynsage

$\mathcal{U}$ and $\mathcal{F}$: the same idea, pushed further

The notebook stops at two loops. Here are bigger diagrams, each done **three independent ways** inside feynsage, and then by FeynCalc's FCFeynmanPrepare:

diagram	L	N	trees	terms of $\mathcal{F}$	trees = $\det M$	Kirchhoff = trees	= FeynCalc	time (trees)
kite	2	5	8	8	✓	✓	✓	< 10 ms
double box	2	7	15	11	✓	✓	✓	< 10 ms
banana (3 loops)	3	4	4	13	✓	✓	✓	< 10 ms
triple box (3 loops)	3	10	56	65	✓	✓	✓	10 ms

```
g = graph("T1-T2, T2-T3, T3-T4, B1-B2, B2-B3, B3-B4, T1-B1, T2-B2, T3-B3, T4-B4",
          {"B1": "p1", "T1": "p2", "T4": "p3", "B4": "-p1 - p2 - p3"}, kin=box_kin)
g.U(), g.F(), g.U_kirchhoff() # trees, 2-forests, matrix-tree theorem
props, loops = g.family(); IntegralFamily("f", loops, g.kin, props).UF() # det M
```

kite: $\mathcal{U} =$

$$x_1x_3 + x_2x_3 + x_1x_4 + x_2x_4 + x_1x_5 + x_2x_5 + x_3x_5 + x_4x_5$$

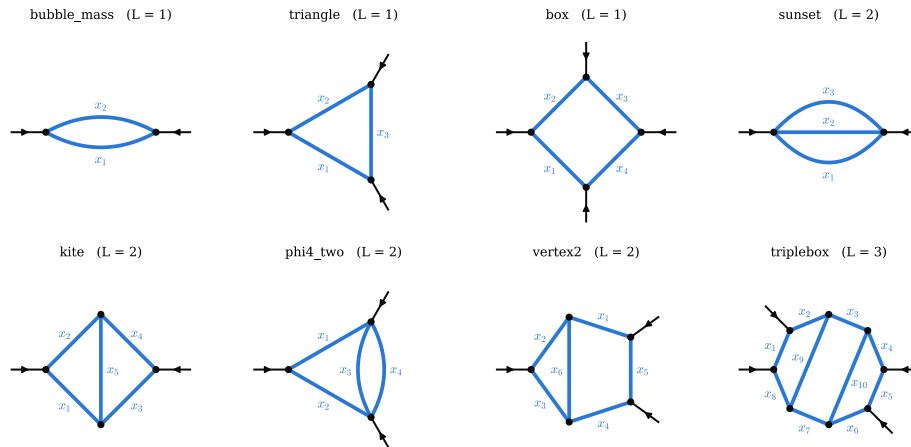
$\mathcal{F} =$

$$-sx_1x_2x_3 - sx_1x_2x_4 - sx_1x_3x_4 - sx_2x_3x_4 - sx_1x_2x_5 - sx_2x_3x_5 - sx_1x_4x_5 - sx_3x_4x_5$$

Graphs: a diagram in one line

```
g = graph("A-B:m, A-B:m, A-B:m",
        {"A": "p", "B": "-p"},
        kin={"p^2": "s"})
g.L, g.U(), g.F(), g.plot()
diagram("triplebox")
```

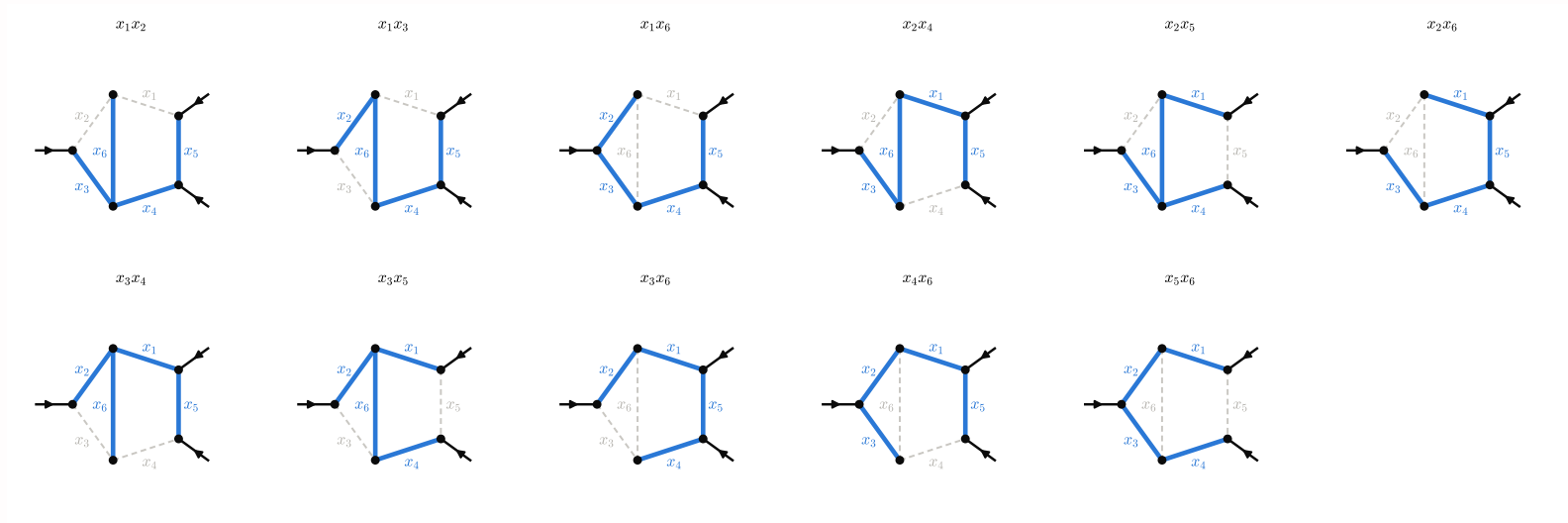
- lines "U-V" or "U-V:mass", then the momentum entering each vertex and the scalar products
- `diagram()` lists the ready-made graphs
- the order of the lines is the numbering $x_1, x_2, \dots$ of $\mathcal{U}$ and $\mathcal{F}$
- massive lines thick and blue, x_i written on each line
- vertices placed with the fewest crossing lines
- the same drawing code makes every picture on the next slides



eight of the twelve diagrams in `diagram()`, drawn by `g.plot()`

Spanning trees, drawn: the two-loop vertex of the note

```
v = diagram("vertex2"); v.spanning_trees(); draw_panels(v, v.spanning_trees())
```

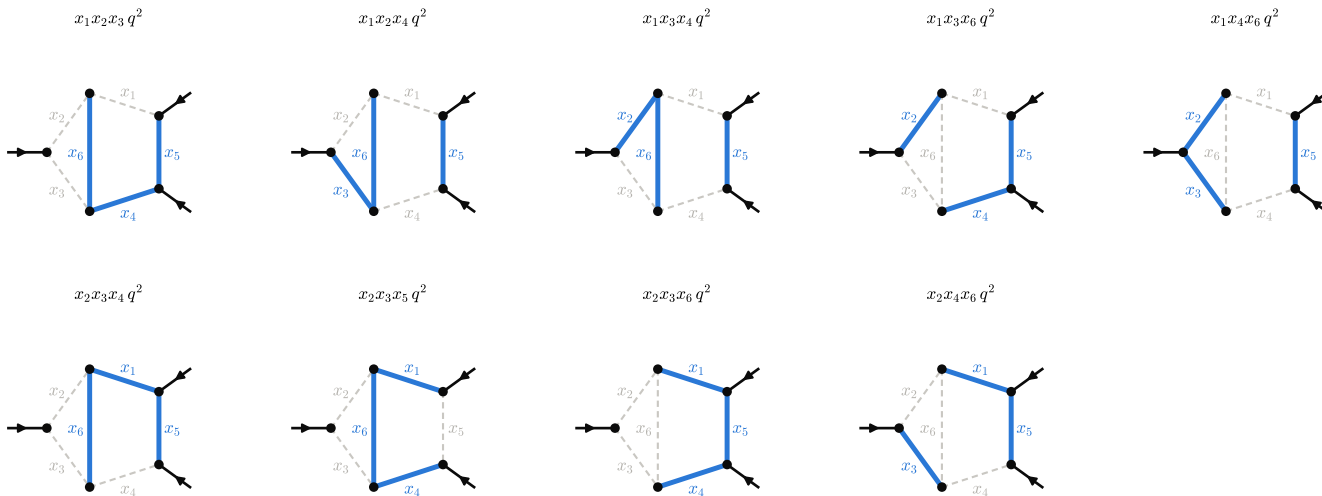


All 11 spanning trees (dashed = removed lines). Each gives the product of its removed x_i :

$$\mathcal{U} = (x_2 + x_3)(x_1 + x_4 + x_5) + (x_1 + \dots + x_5)x_6$$

The same as the lecture note (sector decomposition, Example 3) ✓, and the same as Kirchhoff's determinant and $\det M$ ✓

2-forests that carry momentum



Of the 19 2-forests only 9 separate the q vertex from both on-shell legs; the others carry $p_1^2 = 0$ or $p_2^2 = 0$:

$$\mathcal{F} = -q^2[x_1x_3x_4 + x_1x_2(x_3 + x_4) + x_2x_3(x_4 + x_5) + (x_1 + x_2)(x_3 + x_4)x_6]$$

`v.two_forests()` lists (removed lines, vertices of one tree); `F()` adds up (product of removed x_i) $\times P^2$, then $+\mathcal{U} \sum x_i m_i^2$.

From the graph to an integral family

Momentum routing. `g.family()` takes a spanning tree; every line outside it (a chord) gets a loop momentum, closed through the tree; the external momenta flow through the tree.

```
D1 = (l1)^2
D2 = (-l1 + p)^2
D3 = (l2)^2
D4 = (-l2 + p)^2
D5 = (-l1 + l2)^2
```

Symmetries of the whole family.

Permutations of the lines that leave $\mathcal{U} + \mathcal{F}$ unchanged (Pak's criterion), found as automorphisms of a coloured graph: $[(1, 0, 3, 2, 4), (2, 3, 0, 1, 4)]$

NEW: SECTOR SYMMETRIES

A **sector** = the lines present in an integral (tutorial PDF, Chapter 5). Some symmetries hold inside one sector only: feynsage compares the sector polynomials $(\mathcal{U} + \mathcal{F})|_{x_j=0}$ in canonical form and, with numerators, shifts the loop momenta, as LiteRed and Kira do.

two-loop sunset	before	now
three masses	7	7
equal masses	6	3

Same masters and identical coefficients as Kira 3.1.

How many masters, before any reduction?

Lee and Pomeransky (2013): in each sector set the missing $x_j = 0$ and count the points where every derivative of

$$G = \mathcal{U} + \mathcal{F}$$

vanishes, all remaining $x_i \neq 0$. That is the number of masters of the sector. A Gröbner basis does the counting:

```
I = R.ideal([G.derivative(x) for x in live] +  
[t*prod(live) - 1])  
I.vector_space_dimension()
```

family	critical points	IBP masters
bubble, equal masses	2	2 ✓
bubble, two masses	3	3 ✓
massless box	3	3 ✓
sunset, three masses	7	7 ✓

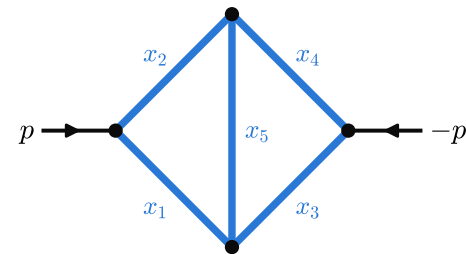
LIMITS OF THE COUNT

- it does not use symmetries: equal-mass sunset gives 4 points in the sunset sector, IBP needs 2
- the points must be isolated; in the massless kite's top sector they form a curve, then one needs the Euler characteristic (Bitoun, Bogner, Klausen, Panzer 2019)

Two loops: IBP reduction of the kite

```
kite = family(["l1", "l1 + q", "l1 + l2", "l1 + l2 + q", "l2"],
kin={"q^2": 1})
ibp_reduce(kite, ["F(2,2,1,2,2)"])
```

$$F(2,2,1,2,2) = \frac{(-4) \cdot (d-9) \cdot (d-6) \cdot (d-5) \cdot (d-3)}{(d-8)} F(1,1,1,1,0) \\ + \frac{(-729) \cdot (d-7) \cdot (d-\frac{16}{3}) \cdot (d-\frac{14}{3}) \cdot (d-\frac{10}{3}) \cdot (d-\frac{8}{3}) \cdot (d^2-11d+20)}{(d-10) \cdot (d-4) \cdot (d-8)^2} F(0,1,1,0,1)$$



Masters: two bubbles $F(1,1,1,1,0)$ and the sunset $F(0,1,1,0,1)$.
Identical to LiteRed and Kira 3.1.

kite, seeds up to	equations	exact, all equations	exact, trimmed	finite fields	Kira 3.1
4 dots, 2 numerators	9 368	37 s	0.6 s	0.7 s	
9 dots, 4 numerators	157 843	11 min	8.6 s	11.7 s	6.7 s*

All on one server (hercules). Trimmed: exact Laporta on only the 859 equations the target needs (method="trimmed"); the three feynsage columns agree exactly. *Kira: another server, one thread.

One loop, the Package-X way

The rank-2 bubble with all symbols, 0.03 s:

```
loop("l^mu l^nu", ["l", "m"], ["l + p", "m"], kin={"p^2": "s"})
```

$$g^{\mu\nu}: \quad \frac{1}{6}m^2 \left(\frac{1}{\epsilon} + \log\left(\frac{\mu^2}{m^2}\right) + 1 \right) + \frac{1}{3}m^2 + \frac{1}{12}(4m^2 - s) \left(\frac{1}{\epsilon} + \Lambda(s; m, m) + \log\left(\frac{\mu^2}{m^2}\right) + 2 \right) - \frac{1}{18}s$$

$$p^\mu p^\nu: \quad \frac{m^2 \left(\frac{1}{\epsilon} + \log\left(\frac{\mu^2}{m^2}\right) + 1 \right)}{3s} - \frac{m^2}{9s} - \frac{(m^2 - s) \left(\frac{1}{\epsilon} + \Lambda(s; m, m) + \log\left(\frac{\mu^2}{m^2}\right) + 2 \right)}{3s} - \frac{4m^2 - s}{18s}$$

$\Lambda(s; m, m)$ is Package-X's DiscB.

- projection on a symmetric basis, Gram matrix inverted **exactly** (adjugate)
- rational terms from $D = 4 - 2\epsilon$, kept to $\mathcal{O}(\epsilon^2)$ when IR poles appear
- DiscB, LogM carry the $+i\epsilon$ branch, as in Package-X
- PVB, PVC, PVD: Package-X's coefficient functions
- Euclidean or Minkowski: euclidean=True
- numerators from FORM traces go straight in

C_0 and D_0 in closed form

`explicit(C0(0, 0, 5, 1, 1, 1))`, exact, with the branches fixed by the $+i\epsilon$:

$$\begin{aligned} & -\frac{2}{15}\pi^2 + \frac{1}{5}i\pi \log\left(\frac{10}{\sqrt{5}+5}\right) + \frac{1}{10} \log\left(\frac{10}{\sqrt{5}+5}\right)^2 \\ & -\frac{1}{5}i\pi \log\left(-\frac{10}{\sqrt{5}-5}\right) + \frac{1}{10} \log\left(-\frac{10}{\sqrt{5}-5}\right)^2 + \frac{1}{5}\text{Li}_2\left(\frac{1}{10}\sqrt{5} + \frac{1}{2}\right) \\ & + \frac{1}{5}\text{Li}_2\left(-\frac{1}{10}\sqrt{5} + \frac{1}{2}\right) \end{aligned}$$

`.n(digits=30)` gives `-0.8943345118780581048846775 - 0.6047086137711147833763413 i`

D_0 : Denner's 16 dilogarithms (no massless line), Denner–Dittmaier for one to four massless lines. `explicit(D0(1,2,3,4,-5,-6,1,1,1,1))` is exact, 8852 characters long, value `0.27279190436796813983`.

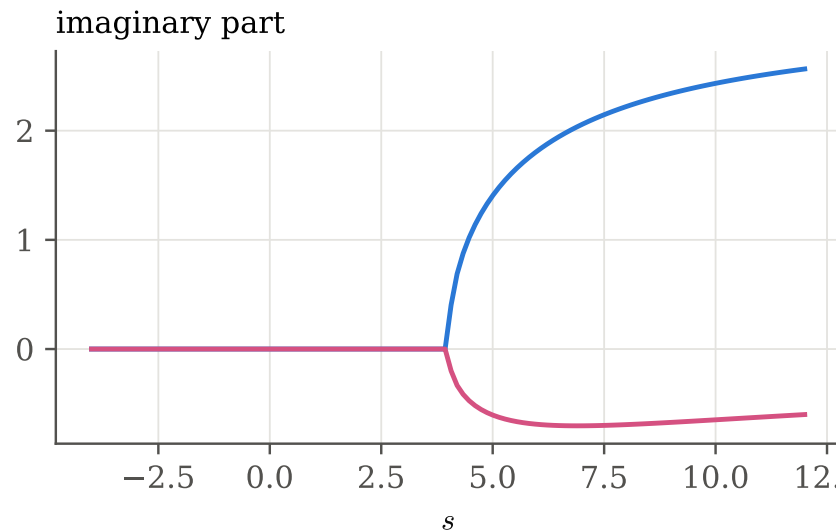
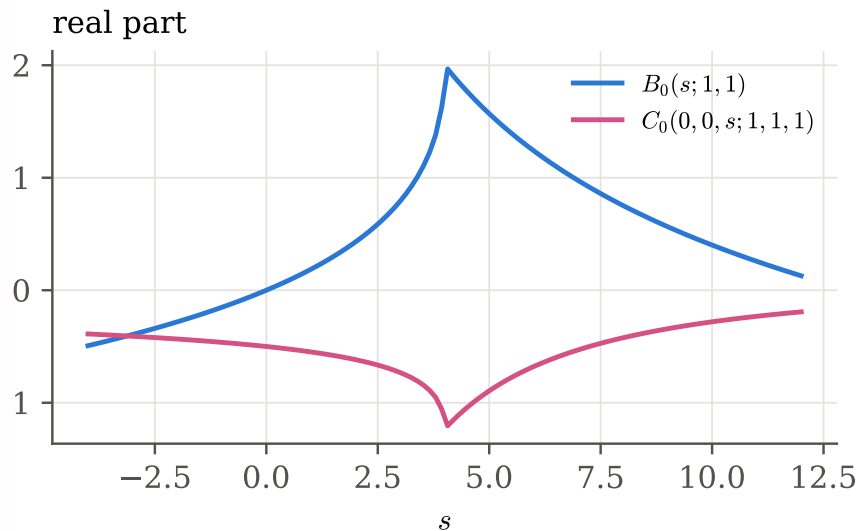
HOW EVERY BRANCH IS FIXED

Each quantity is carried twice: as an 80-digit number with a tiny $+i\epsilon$ and as an exact expression. Where an argument lies on a cut, the number decides the side. At the end the two are compared at two precisions.

Through the thresholds

```
quick_plot([B0(s, 1, 1), C0(0, 0, s, 1, 1, 1)], (s, -4, 12), parts="both")
```

finite part (ϵ^0) at $\mu=1$



Soft and collinear poles

All 6 triangles and 16 boxes of Ellis and Zanderighi, any order of the lines, in the Package-X normalisation:

$C_0(0, 5, 0, 0, 0, 0)$ (massless, $s = 5$, $\mu = 1$):

$$\frac{1}{5\epsilon^2} + \frac{1}{\epsilon} \left(\frac{1}{5}i\pi - \frac{1}{5}\log(5) \right) - \frac{7}{60}\pi^2 - \frac{1}{5}i\pi\log(5) + \frac{1}{10}\log(5)^2$$

$D_0(0,0,0,0, -2,-3, 0,0,0,0)$ (massless box, $\mu = 1$):

$$\frac{2}{3\epsilon^2} + \frac{1}{\epsilon} \left(-\frac{1}{3}\log(3) - \frac{1}{3}\log(2) \right) - \frac{2}{9}\pi^2 + \frac{1}{3}\log(3)\log(2)$$

QED vertex $C_0(m^2, q^2, m^2; 0, m, m)$, photon massless, electrons on shell ($m = 1$, $q^2 = -3$): $1/\epsilon$ coefficient 0.3419036239, finite part -0.1252442693 .

CHECKED

118 points against Package-X (all three orders in ϵ , double precision). Boxes 15 and 16, whose finite parts Package-X cannot evaluate, against a photon mass λ : $\log \lambda^2 \rightarrow 1/\epsilon + \log \mu^2$.

When the Gram determinant vanishes

The $g - 2$ vertex at $q^2 = 0$: $p_1 = p_2$, so the usual reduction divides by zero. feynsage takes the coefficients straight from Feynman parameters (lines with identical kinematics are grouped and integrated out exactly):

```
PVC(0, 1, 0, m^2, 0, m^2, 0, m, m)
PVC(1, 0, 0, m^2, 0, m^2, 0, m, m)
```

$$C_1 = \frac{1}{2m^2}$$

$$C_{00} = \frac{1}{4\epsilon} - \frac{1}{2} \log(m) + \frac{1}{2} \log(\mu) + \frac{1}{4}$$

PACKAGE-X

$$C_1 = 1/(2m^2), \quad C_{00} = \frac{1}{4} + \frac{1}{4} \left(\frac{1}{\epsilon} + \log \frac{\mu^2}{m^2} \right) \quad \checkmark$$

Also all invariants zero (divided differences, any masses) and a massive photon: 11 more points agree with Package-X.

Real physics, computed

QED, PESKIN AND SCHROEDER

- vacuum polarisation: Ward identity $C_g + q^2 C_{qq} = 0$ **exactly**
- Peskin (7.91) reproduced to 10^{-15}
- running of α from the leptons at one loop: $\Delta\alpha(M_Z^2) = 0.031419$
- electron $g - 2$: $\mathcal{U}, \mathcal{F}$ from the vertex graph, Dirac algebra in FORM, $F_2(0) = \alpha/(2\pi)$ **exactly**

$\pi^0 \rightarrow \gamma\gamma$ FROM THE QUARK TRIANGLE

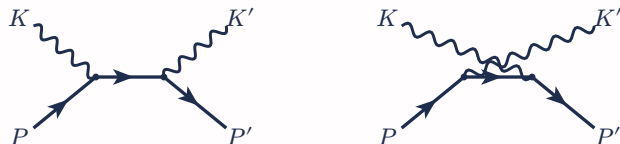
- trace with γ_5 in FORM:
 $4im \varepsilon(k_1, k_2, \mu, \nu)$
- triangle from its graph polynomials
- width 7.79 eV vs PrimEx-II 7.80 ± 0.12 eV
- lifetime 8.35×10^{-17} s vs PDG 8.43×10^{-17} s

```
T = form.dirac_trace(["mu", "l + m", "nu", "l + q + m"], vectors=["l", "q"], dim="D") # FORM
r = loop(form.to_loop(T).replace("D", "d"), ["l", "m"], ["l + q", "m"], kin={"q^2": "s"})
(r["g^{mu nu}"] + s * r["q^mu q^nu"]).expand() # 0: the Ward identity, exactly
```

Worked out in `examples/peskin_examples.ipynb` and `examples/feynsage_walkthrough.ipynb`.

From Chluba's thesis (2005): Compton scattering

J. Chluba, *Spectral distortions of the cosmic microwave background* (2005): Compton, double Compton and their infrared divergence. Every QED result there was recomputed twice, with feynsage + FORM and with FeynCalc.



BOTH PROGRAMS GIVE, EXACTLY

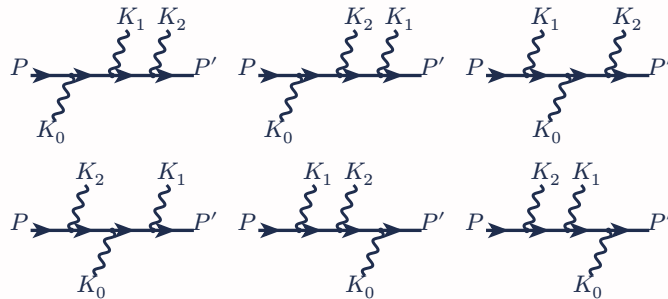
$$\sum |\mathcal{M}|^2 = 8e^4 \left[-\frac{u - m^2}{s - m^2} - \frac{s - m^2}{u - m^2} + 4m^2 \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) + 4m^4 \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right)^2 \right]$$

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2}{2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \right) \quad (\text{Klein-Nishina}), \quad \sigma_T = \frac{8\pi}{3} r_0^2 \quad (\text{Thomson limit})$$

FeynCalc: 0.2 s, symbolic. feynsage: exact rationals through FORM. Script: `examples/chluba/`.

Double Compton: six diagrams

$e(P) + \gamma(K_0) \rightarrow e(P') + \gamma(K_1) + \gamma(K_2)$: the photons attach to the electron line in all $3! = 6$ orders.



AGAINST MANDL & SKYRME'S X (THESIS EQ. D.1)

At four random rational points both programs give **exactly** $\sum_{\text{spins, pol.}} |\mathcal{M}|^2 = 4e^6 X$: the thesis's $|\mathcal{M}|^2 = e^6 X$ is the average over the 2×2 initial states.

36 traces, 12 γ 's time per point

feynsage + FORM	0.5 s
FeynCalc	7.7 s

The code: Compton and double Compton

feynsage + FORM examples/chluba/compton_dc.sage FeynCalc examples/chluba/compton_dc_feyncalc.wl

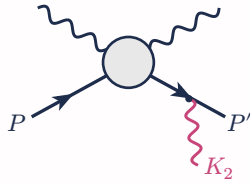
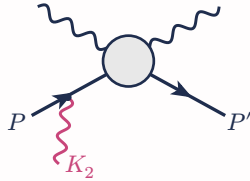
```
from feynsage import form
# Compton: (chain, conjugate chain, s-m^2 or u-m^2)
ch = [(["nu", "p + k + m", "mu"],
       ["mu", "p + k + m", "nu"], 2*pk),
      (["mu", "p - kp + m", "nu"],
       ["nu", "p - kp + m", "mu"], -2*pk)]
tr = lambda f: evaluate(form.dirac_trace(
    f, vectors=["p", "pp", "k", "kp"]), table)
msq = sum(tr(["pp + m"] + a + ["p + m"] + bc) / (da*db)
          for a, _, da in ch for _, bc, db in ch)

# double Compton: photons as outgoing momenta
ph = {"a": ("k0", "mu1"), "b": ("k1", "mu2"),
      "c": ("k2", "mu3")}
for o in itertools.permutations("abc"): # 6 diagrams
    (q1, i1), (q2, i2), (q3, i3) = (ph[x] for x in o)
    chain = [i1, "pp + %s + m" % q1, i2,
              "p - (%s) + m" % q3, i3]
# msq = -(sum of the 36 traces / denominators)
msq / mandl_skyrme_X(table) # = 4, exactly
```

```
chain = GA[nu].(GS[p+k]+m).GA[mu]/(s-m^2) +
          GA[mu].(GS[p-kp]+m).GA[nu]/(u-m^2);
chainbar = GA[mu].(GS[p+k]+m).GA[nu]/(s-m^2) +
            GA[nu].(GS[p-kp]+m).GA[mu]/(u-m^2);
msq = TR[(GS[p+k-kp]+m).chain.(GS[p]+m).chainbar]
      // Contract // ExpandScalarProduct // Simplify

(* double Compton: the 6 orders of the photons *)
ph = {{-k0, i1}, {k1, i2}, {k2, i3}};
chains = Map[
  {GA[#[[1,2]]].(GS[pp + #[[1,1]] + 1).GA[#[[2,2]]].
   (GS[p - #[[3,1]] + 1).GA[#[[3,2]]], ...} &,
  Permutations[ph]];
msq = -ExpandScalarProduct[Contract[
  TR[(GS[pp]+1).M.(GS[p]+1).Mbar]]];
msq/mandlSkyrme[...] (* = 4, exactly *)
```

The infrared divergence (thesis sec. 4.4.5)



a soft photon K_2 from an external electron

For $K_2 \rightarrow 0$ double Compton factorises into Compton times the **eikonal factor**:

$$\sum |\mathcal{M}_{\text{DC}}|^2 \rightarrow e^2 S(K_2) \sum |\mathcal{M}_{\text{C}}|^2$$

$$S = \frac{2P \cdot P'}{(P \cdot K_2)(P' \cdot K_2)} - \frac{m^2}{(P \cdot K_2)^2} - \frac{m^2}{(P' \cdot K_2)^2}$$

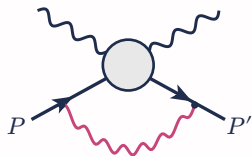
K_2 scaled by	10^{-4}	10^{-6}	10^{-8}
ratio of the two sides	0.9969856372	0.9999698113	0.9999996981

$S \propto 1/\omega_2^2$, so $d\sigma \propto d\omega_2/\omega_2$: the divergence of the thesis. Integrated over angles, a soft photon is emitted with $dN = \alpha/\pi I(t) d\omega_2/\omega_2$, $I(t) = 2P \cdot P' \int_0^1 \frac{dx}{m^2 - x(1-x)t} - 2$.

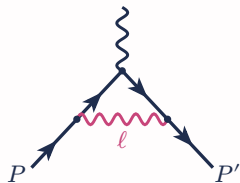
LIGHTMAN'S LAW (THEESIS EQ. 4.24), DERIVED

For cold electrons and soft photons $I(t) = -2t/3 + \dots$; averaged over Thomson scattering, $-t = 2\omega_0^2(1 - \cos \theta) \rightarrow 2\omega_0^2$, so $dN/d\omega_2 = \frac{4\alpha}{3\pi} \omega_0^2/\omega_2$ per Compton scattering: exactly the $4\alpha/3\pi$ of the thesis.

Regularisation: the divergence cancels



the virtual soft photon between the electron legs



its pole is that of the QED vertex at $t = (P - P')^2$

The thesis stops the divergence with a lowest frequency $\nu_{2,\min}$.
In $D = 4 - 2\varepsilon$ it cancels:

$$\text{real } (\omega_2 < \Delta E) : -\frac{\alpha I(t)}{2\pi\varepsilon}, \quad \text{virtual } (2 \operatorname{Re} F_1) : +\frac{\alpha I(t)}{2\pi\varepsilon}$$

The vertex, computed with the loop in D dimensions, its IR pole from the soft triangle $C_0(m^2, t, m^2; 0, m, m)$ (UV poles do not depend on t , so differences isolate the IR part):

t	$I(t) - I(-1/2)$	feynsage	FeynCalc + Package-X
-1	0.271963043918	0.271963043918	0.271963043918
-3	1.108545637288	1.108545637288	1.108545637288
-8	2.368915948651	2.368915948651	2.368915948651

Normalisation check: $F_2 \rightarrow \alpha/2\pi$ as $t \rightarrow 0$ in both (1.99999967 at $t = -10^{-6}$, in units $\alpha/4\pi$).

so

the cutoff $\nu_{2,\min}$ of the thesis becomes $\log(\Delta E)$, the energy resolution: Bloch–Nordsieck, computed.

The code: the vertex in D dimensions

feynsage + FORM examples/chluba/
ir_cancellation.sage

```
# the vertex projected on gamma^mu (and on (P+P')^mu)
num = form.dirac_trace(
    ["p + m", "mu", "pp + m", "nu",
     "pp + l + m", "mu", "p + l + m", "nu"],
    vectors=["p", "pp", "l"], dim="D")
r = loop(form.to_loop(num).replace("D", "d"),
    [{"l", "0"}, {"l + pp", "1"}, {"l + p", "1"}],
    kin={"p^2": 1, "pp^2": 1, "p.pp": 1 - t/2})
# r["1"] holds A0, B0 and the soft triangle
# C0(1, t, 1; 0, 1, 1) with its IR pole
A, B = Mx.solve_right(vector(SR, vs)) # Gordon basis
F1, F2 = A + B, -B # units alpha/4pi
F1.coefficient(eps, -1) # = I(t) + const: IR pole
```

FeynCalc + Package-X

ir_cancellation_feyncalc.wl

```
numG = DiracTrace[(GSD[p]+1).GAD[mu].(GSD[pp]+1).
    GAD[nu].(GSD[pp+l]+1).GAD[mu].(GSD[p+l]+1).GAD[nu],
    DiracTraceEvaluate -> True];
loopG = PaXEvaluate[
    ToPaVe[TID[ExpandScalarProduct[Contract[numG]] *
        FAD[{l,0}, {l+pp,1}, {l+p,1}], l], l]/(I Pi^2),
    PaXImplicitPrefactor -> 1];
AB = LinearSolve[M, {loopG, loopP}];
F1 = Normal[Series[AB[[1]] + AB[[2]],
    {Epsilon, 0, 0}]];
Coefficient[F1, Epsilon, -1] (* the same pole *)
```

Run time for four values of t : feynsage 1.3 s, FeynCalc + Package-X 6.6 s.

Comparison with other packages

	feynsage	Package-X	LoopTools	FeynCalc	LiteRed	FIRE	Kira
needs Mathematica	no	yes	no	yes	yes	yes ^a	no
one-loop tensor reduction	✓	✓	numbers	✓			
C_0 as a formula	✓	✓		via Package-X			
D_0 as a formula	✓	IR cases	numbers	via Package-X			
IR poles of C_0 , D_0	✓	✓	✓	via Package-X			
multi-loop IBP	✓				✓	✓	✓
finite fields	✓					✓	✓
$\mathcal{U}$, $\mathcal{F}$ from the graph	✓			✓			
Dirac algebra	FORM	Spur		✓			

An empty cell: we did not find this feature in that program. ^aFIRE has a C++ core and a Mathematica front end.

Head to head: the same job, measured

Speed (same laptop, symbols everywhere)

task	feynsage	Package-X
C_{001}	0.42 s	0.95 s
C_{112}	0.39 s	1.59 s
D_{00}	6.0 s	0.08 s
10 numerical D_0	0.23 s	0.25 s
digits of a numerical D_0	≈ 40	16

Package-X is much faster for symbolic box tensors: honest numbers.

Agreement

against	result
Package-X, 190 points	every digit of its values
Package-X, IR, 118 points	double precision
LoopTools 2.16 (Denner), hard D_0 points	10^{-15}
Kira 3.1, LiteRed	identical coefficients
FeynCalc, $\mathcal{U}$ and $\mathcal{F}$	identical polynomials
D_0 relabellings	all 24 at 15 points; 6 random ones at 2000 random points

A point where Package-X and LoopTools fail

$C_0(-4, -3, -1; 0, 0, 1)$: every invariant is spacelike, so $C_0 = -\int d^3x \delta(1 - \sum x_i)/\mathcal{F}$ must be **real and negative**.

program	value	right?
feynsage (closed form)	-0.585976809672364723	✓
Mathematica NIntegrate (no Package-X)	-0.585976809672364723	✓
FeynCalc FCFeynmanParametrize , integrated	-0.58597680967236472	✓
mpmath, 45 digits	-0.585976809672364722650390572218069267	✓
Package-X 2.1.1, two labellings	+0.671253, -0.721045	✗
LoopTools 2.16, version a (FF)	+0.671253, -0.721045, -1.843207	✗
LoopTools 2.16, version b (Denner)	four different complex numbers	✗

Red = wrong. The true value is $-0.5859768096\dots$; it cannot change when the lines are relabelled, and it must be real and negative. At nearby points Package-X and LoopTools' FF version agree with **NIntegrate** to 16 digits (the Denner version is still wrong there). In feynsage: `C0(-4, -3, -1, 0, 0, 1).n(digits=40)`. Package-X's wrong values are LoopTools' FF values digit for digit.

How we know it is right

- 12 test files and a random stress test, run on a laptop (Sage 10.9) and on a CentOS 7 server (Sage 10.7)
- every formula checked against another program **or** a direct integral
- D_0 is symmetric under the 24 relabellings of its lines: all 24 agree at 15 points, 6 random ones at each of 2000 random points
- exact expressions are evaluated at two precisions before they are returned
- two independent code audits; every finding fixed and turned into a test

LIMITS, HONESTLY

- finite-field IBP: D and at most one invariant
- closed forms of C_0 , D_0 need numerical kinematics; with symbols they stay symbols
- vanishing Gram determinant: one group of identical lines, or two groups with equal masses
- symbolic box tensors are slower than Package-X

Summary

FEYNSAGE

- graphs $\rightarrow \mathcal{U}$, $\mathcal{F} \rightarrow$ families $\rightarrow$ IBP $\rightarrow$ masters
- one loop: exact tensor reduction, C_0 and D_0 in closed form, IR poles, zero Gram determinants
- QED of Chluba's thesis: Compton, double Compton, the IR divergence cancelled in D dimensions
- free, in SageMath, checked against Package-X, LoopTools, FeynCalc, LiteRed and Kira

```
git clone https://github.com/aburousan/feynsage
cd feynsage && ./install.sh --test
```

Start with the tutorial PDF docs/tutorial/feynsage_tutorial.pdf or the notebooks in tutorials/, then examples/feynsage_walkthrough.ipynb.

Thanks to Prof. B. Ananthanarayan for the lecture and the uf-new-short.nb notebook that started the $\mathcal{U}$, $\mathcal{F}$ part.